INCH-POUND
MIL-DTL-2857C
1 October 2020
SUPERSEDING
MIL-G-2857B
10 November 1988

DETAIL SPECIFICATION

GLASS, HEAT-TREATED, GLAZING, RECTANGULAR (FOR BRIDGE WINDOWS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers rectangular heat-treated quality clear glazing glass for bridge windows. This specification does not include bulletproof glass.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM C1036 - Standard Specification for Flat Glass

(Copies of this document are available online at www.astm.org.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://webstore.ansi.org.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 5620

DISTRIBUTION STATEMENT A. Approved for public release. Distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:49Z
Check the source to verify that this is the current version before use.

3.2 Material. The glass used for heat treatment shall be glazing quality, clear, polished plate glass (except as to nominal thickness and tolerances) in accordance with the requirements of ASTM C1036, type I, class 1, quality Q3.

3.2.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Construction. Heat-treated glass shall be uniformly heated by any standard process capable of producing glass of the specified strength characteristics.

3.4 Tolerance. Unless otherwise specified (see 6.2), all dimensional tolerances (thickness, length, width, and squareness) shall be in accordance with ASTM C1036.

3.5 Finish. The corners shall be slightly rounded, and all edges shall be ground or fire-polished to a neat finish.

3.6 Grooving. The glass shall be properly grooved or beveled, as specified (see 6.2).

3.7 Thickness. The thickness of the glass shall be ¼, 5⁄16, or ⅜ inch, as specified (see 6.2).

3.8 Impact resistance. Glass shall not crack or show any visible change (see 4.6.1).

3.9 Strain pattern. When viewed under polarized light, the glass shall show either concentric colored or black rings, or a uniformly distributed colored or black strain pattern, characteristic of fully heat-treated glass (see 4.6.3).

3.10 Light transmission. The heat-treated plate glass shall show a transmission of white light of not less than 86 percent (see 4.6.2).

3.11 Waviness. The glass shall exhibit no objectionable waves or similar objectionable defects in accordance with ASTM C1036 (see 4.5 and 4.6.4).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on a sample when a first article inspection is required (see 3.1). This inspection shall include the examinations of 4.5 and the tests of 4.6.1 through 4.6.4 (see table I).

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:49Z
Check the source to verify that this is the current version before use.

TABLE I. First article and conformance inspections.

Characteristic	First article	Conformance	Requirement	Verification
Material	X	X	3.2	4.5
Construction	X	X	3.3	4.5
Tolerance	X	X	3.4	4.5
Finish	X	X	3.5	4.5
Grooving	X	X	3.6	4.5
Thickness	X	X	3.7	4.5
Impact resistance	X	-	3.8	4.6.1
Strain pattern	X	-	3.9	4.6.3
Light transmission	X	-	3.10	4.6.2
Waviness	X	X	3.11	4.5 and 4.6.4

4.3 Conformance inspection. Conformance inspection shall include the examinations of 4.5 and the tests of 4.6.4 (see table I).

4.4 Sampling. A random sample of glass panels shall be selected from each lot offered to the Government in accordance with ANSI/ASQ Z1.4 at inspection level II and Acceptance Quality Limit (AQL) as specified (see 6.2).

4.5 Visual and dimensional examinations. Samples shall be examined to determine compliance with 3.2, 3.3, 3.5, and 3.11 for visual examination and 3.4, 3.6, and 3.7 for dimensional examination.

4.6 Tests. A sample (see 4.4) shall be subjected to the tests specified in 4.6.1 through 4.6.4.

4.6.1 Impact test. Each sample item shall be inspected at the place of manufacture for conformance to 3.8. Material selected for the impact test shall be placed on the frame described below and shall be subjected to the impact of a ½-pound steel ball striking a single gravity-accelerated blow at the center of the specimen from a height as shown in table II. The testing frame shall consist of an upper and lower piece of l-inch thick iron plate. The inside dimensions of the rectangular frame shall be 12 by 12 inches and the outside dimensions shall be 16 by 16 inches. The plates shall be cut from a 1-inch thick steel plate. The surface shall be smoothly planed and shall be covered with rubber gasket material ¼ inch in thickness. The lower frame shall be bolted to a wooden frame of such thickness that the clearance between the under glass surface and the floor is 4 inches. Since heat-treated glass sheets cannot be cut to size, it is necessary to use the sheets as received. The lower frame shall be placed on the floor, and the sheet of glass shall be placed so that the “overhang” is nearly uniform. The upper frame shall then be placed with the rubber-protected edge down, so that the inner and outer edges coincide with the lower frame. The ball suspension shall be so placed as to drop the ball as near as possible to the center area of the glass plate bounded by the frame.

TABLE II. Impact test measurements.

Thickness of glass (inch)	Height of fall of ball (feet)
¼	7
5/16	8¼
⅜	10

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:49Z
Check the source to verify that this is the current version before use.

4.6.2 Light transmission test. A sample shall be tested at the place of manufacture. The rectangular (parallel) luminous transmittance of the sheet of glass shall be measured at normal incidence calculated using the equation Transmission factor (T) = Transmitted light/Incident light. The original value of incident light shall be the photometer reading without the glass in position. The light transmission shall be considered as the percentage of the photometer readings with the glass in position as compared to the reading without the glass in position.

4.6.3 Polariscopic examination. A sample shall be subjected to polariscopic examination as specified in 3.9. When ordinary transparent material such as glass is subjected to stresses, the strained regions inside are characterized by changes in their optical properties, whereby they become partially birefringent. The amount of birefringence produced at any point in the material is, in part, a measure of the strain existing there. Consequently, when a sample glass containing strains is inserted between crossed polarizers, light is restored in the region under strain. This restored light usually appears in the form of various colored striations or colored patterns. If any glass does not show concentric rings as specified in 3.9, the lot represented shall be rejected. The strain pattern in properly heat-treated plate glass shall take the form of concentric rings of color near the edge of the plate of glass or may represent regular patterns of color (for example, Scotch plaid). The glass shall be moved to right or left as the pattern may not be visible at the particular angle in which the glass is placed with respect to the viewing polariscope used. A Nicol prism, or equal, or polaroid-equipped lens may be used, so long as the apparatus is equipped with a shutter mechanism carrying a tint plate so that the strain pattern may be viewed alternately as black lines or as a colored-fringed pattern.

4.6.4 Waviness examination. Samples shall be visually examined under an equivalent of light from a clear sunless sky when placed in a vertical position ±10 degrees. Under such conditions, and with the viewer's eye on a level with the center of the sample, the viewer shall look through the sample from a distance of 36 inches, toward a brick wall or similar background at the same distance, showing straight parallel lines. During this examination, the viewer's head shall move horizontally through a slight distance to each side.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The glass covered by this specification is intended to be used for bridge windows aboard naval ships subjected to a combat environment and for general architectural use in construction. The glass covered by this specification is intended to last life of ship. Current stock of glass may be used until depleted; future requisitions for glass should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. When first article is required (see 3.1).
c. Dimensional tolerances if other than specified (see 3.4).
d. Beveling or grooving required (see 3.6).
e. Thickness of glass desired (see 3.7).
f. AQL of 1.0 (see 4.4).
g. Packaging requirements (see 5.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:49Z
Check the source to verify that this is the current version before use.

6.3 Subject term (key word) listing.

Grooving

Impact resistance

Light transmission

Polariscopic examination

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – CR4
Navy – SH

Preparing activity:
Navy – SH
(Project 5620-2020-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:49Z
Check the source to verify that this is the current version before use.